Deep Learning

* Deal with unstucture data (Images, Audio, video).
* It have Inbuild feature Engineering Techinque.
* Neuron's is the base unit of deep Leaning.

⇒ DL is the Subset of ML that uses Artifical Neural Network with many layers to learn Pattern from data.

Neuron

Σ = Summation
θ = Activation.

Neuron.

Building Multiple neuron is known as Neuron architecture.

Artificial Neuron is also known as "Perception"

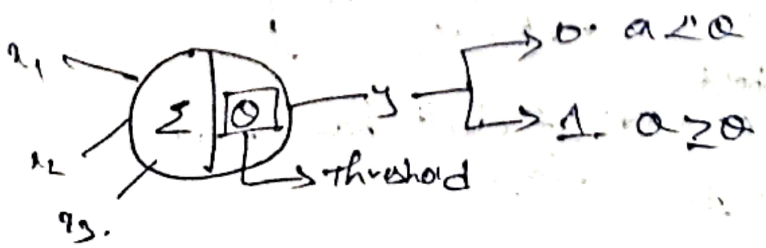
Use Cases

* Image Recognition
* Speech Recognition
* Natural language Processing (NLP)

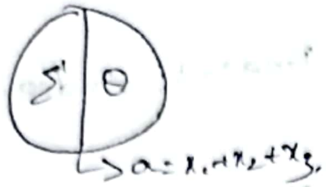
Etc.;

History

MC Culloch Pitts in 1940's



* the output Can only be in binary.



Drawbacks

1. Cannot real time dataset
2. Equal weights for all features.
3. Threshold has been choosen by us.

Based on more disadvantages. So we move onto "SLP" (Single layer Perceptron)

Single layer Perceptron (SLP)

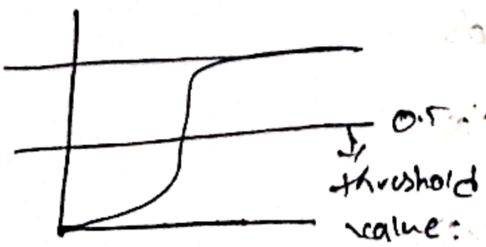
* Simplest type of ANN.
* It Contains only one layer of trainable weights and it mainly used for binary Classification problem.

Note

If ~~the~~ the output is binary classification we ch Sigmoid. for regression we choose Linear.

disadvantage

A Single SLP Can only handle linear handles. Can't able to solve Non linear boundaries.



threshold value formula $= \frac{e^x}{1+e^x}$

* It don't have hidden layers.

If 0.5 more than 1 or less than 0.

Based on some disadvantages. we moving into "MLP" Multi layer Perceptron.

Multi layer Perceptron (MLP)

is a type of ANN that Contains multiple layer of neurons.

In MLP it have Contineous learning process so it is Known as "epoch"

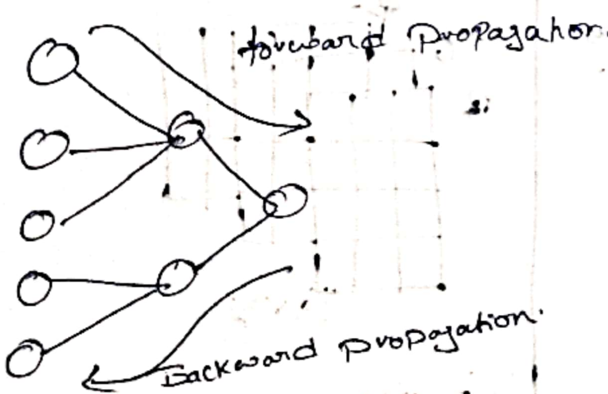
* Majorly use only in supervised.

there are two main process used to train a Neural Network in MLP.

* forward Propagation.
* Backward Propagation.

⇒ forward Propagation is process where input data moves forward through the Neural network to Produce a Prediction.

⇒ Backward Propagation is process where is sent backward through the network to update the weight and bias.



If a Error detector in forward Propagation we need to do Backward Propagation

How the forward and backward works.

Input data
↓
forward Prop
↓
Prediction
↓
Loss Calculation
↓
Backward Prop
↓
weight update
↓
Repeat for many Epochs

Image

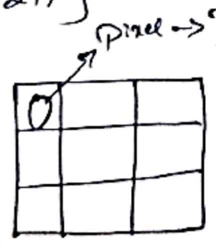
Black & white (Gray scale)

0 → Black
255 → White.

[0-255]

pixel → Brightness value.

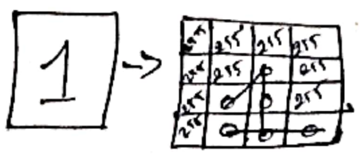
Q1



3x3.

[3x3x1]

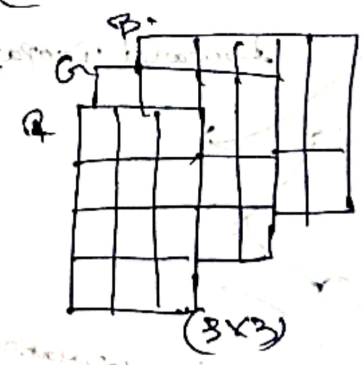
rows X Column X No of channel.



Coloured Image.

(0-255) R (160)
(0-255) G (50)
(0-255) B (50)

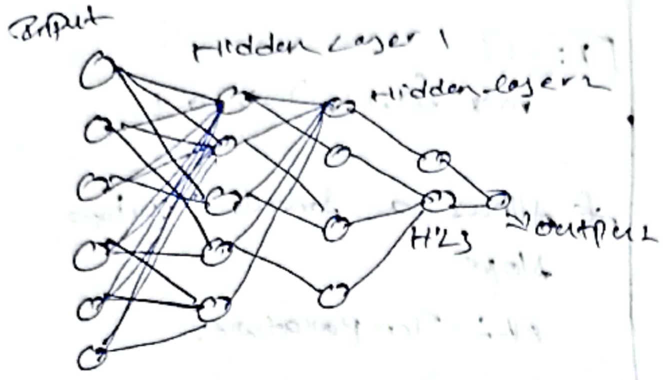
→ Orange



(3x3x3)

Structure

* Input layer
* Hidden layer
* output layer



In Input layer are known as (relay). It doesn't have Activation & summation.

things in our hand ; **By model**

things in our hand	By model
① Network Structure	① Weights
②. Activation function	② Bias
③ learning rate	
④ No. of epochs	
⑤ Batch size	
⑥ optimizer	

Activation function

Is a mathematical function used in a neuron to transform the input signal into an output signal.

→ adds non-Linearity

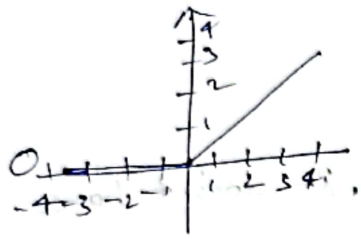
Activation fn in Hidden Layers :-

Relu

Max (0,x)

→ taken +ve → as such
-ve → 0

x=5 max(0,5) = 5
-5 max(0,-5) = 0



* Cuts of negative values completely

Leaky Relu.

max (0·1, x)

$\boxed{x = -1}$
max (0.1, -1) → -0.1

* Allows a small negative slope

Ex:- Temperature.

Activation fn in output layers

Binary classification → Sigmoid
Regression → linear
Multilabel classification → Softmax

Softmax

for softmax we need to do OneHot Encoding.

* Softmax Converts a Set of Values into Probability.

formula :- $\dfrac{e^{x_i}}{\Sigma e^{x_i}}$

Tensorflow is the library of ML Library
Keras is an Api

Tensorflow
→ open source ML Tool (By google)
→ DL Tools.

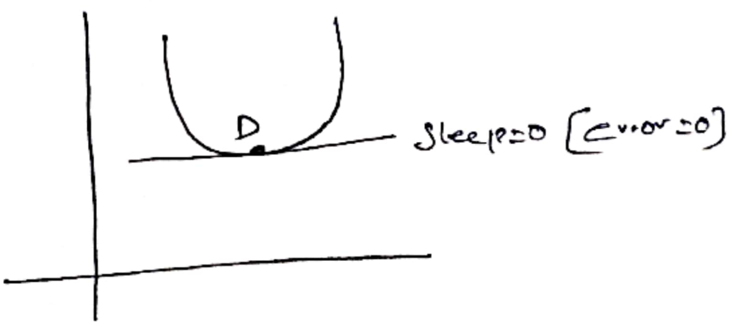
Keras → High level API
which simplifies & train NN.

Optimizers

① Gradient decent

Work flow

① Randomly assigns weights & bias
② check error
③ Reassigns weight & bias.
④.



(2) Momentum in GD

Advanced version of GD.

③ RMS Prop

Ability to adjust learning rate(α).

④ Adam optimizer

It works like both momentum and Rmsprop

Errors

for Regression

* MAE
* MSE
* RMSE

for Classification

* Binary cross entropy.
  [formula = $-y \times \log \hat{y} - (1-y)\log(1-\hat{y})$]
* Categorical Cross entropy (Multi label).
  [formula = $-\sum a \times \log(p)$].

Note.

If we didn't done onehot Encoding then we need to use Sparse_Categorical cross entropy.